

## **Exercise 2: Stored Procedures**

### **Aim**

To create and execute Stored Procedures in MySQL for retrieving and inserting employee details.

### **Theory**

#### **What is a Stored Procedure?**

A **Stored Procedure** is a collection of SQL statements stored in the database that can be executed whenever required. It helps automate repetitive database operations.

#### **Advantages of Stored Procedures**

- Improves query performance.
- Reduces code duplication.
- Enhances database security.
- Makes applications easier to maintain.
- Allows parameter passing for dynamic queries.

#### **Parameters in Stored Procedures**

- **IN** – Accepts input values.
- **OUT** – Returns values to the caller.
- **INOUT** – Works as both input and output.

### **Syntax**

```
CREATE PROCEDURE ProcedureName()
```

```
BEGIN
```

```
    SQL Statements;
```

```
END;
```

### **Applications**

- Inserting records.
- Retrieving records.
- Updating records.
- Deleting records.

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

|   | EmployeeID | FirstName | LastName | DepartmentID | Salary   | JoinDate   |
|---|------------|-----------|----------|--------------|----------|------------|
| ▶ | 1          | Ravi      | Kumar    | 101          | 50000.00 | 2023-01-15 |
|   | 2          | Priya     | Sharma   | 102          | 60000.00 | 2022-05-10 |
|   | 3          | Rahul     | Reddy    | 101          | 55000.00 | 2024-02-20 |
|   | 4          | Anitha    | Rao      | 103          | 65000.00 | 2025-06-20 |
| ● | NULL       | NULL      | NULL     | NULL         | NULL     | NULL       |

Result Grid

Form Editor

Field

### Create a Stored Procedure to Retrieve Employees by Department :

```
--  
3 DELIMITER //  
4  
5 • CREATE PROCEDURE GetEmployeeByDepartment(IN DeptID INT)  
6 BEGIN  
7     SELECT *  
8     FROM Employees  
9     WHERE DepartmentID = DeptID;  
10 END //  
11  
12 DELIMITER ;  
13
```

### Execute the Stored Procedure :

```
CALL GetEmployeeByDepartment(101);
```

```
DELIMITER //
```

### OUTPUT :

| Result Grid  |            |           |          |                    |          |            |
|--------------|------------|-----------|----------|--------------------|----------|------------|
| Filter Rows: |            | Export:   |          | Wrap Cell Content: |          |            |
|              | EmployeeID | FirstName | LastName | DepartmentID       | Salary   | JoinDate   |
| ▶            | 1          | Ravi      | Kumar    | 101                | 50000.00 | 2023-01-15 |
|              | 3          | Rahul     | Reddy    | 101                | 55000.00 | 2024-02-20 |

### Create the sp\_InsertEmployee Stored Procedure :

```
7  
8 • CREATE PROCEDURE sp_InsertEmployee(  
9     IN First_Name VARCHAR(50),  
10    IN Last_Name VARCHAR(50),  
11    IN Department_ID INT,  
12    IN Salary_Value DECIMAL(10,2),  
13    IN Join_Date DATE  
14 )  
15 BEGIN  
16     INSERT INTO Employees  
17     (FirstName, LastName, DepartmentID, Salary, JoinDate)  
18     VALUES  
19     (First_Name, Last_Name, Department_ID, Salary_Value, Join_Date);  
20 END //  
21  
22 DELIMITER ;  
23
```

## Execute the Insert Procedure :

- ```
CALL sp_InsertEmployee(  
    'Anitha',  
    'Rao',  
    103,  
    65000,  
    '2025-06-20'  
);
```
- ```
SELECT * FROM Employees;
```

## Output :

| Result Grid                             |            |           |          |              |          |            |
|-----------------------------------------|------------|-----------|----------|--------------|----------|------------|
| Filter Rows:                            |            |           |          |              |          |            |
| Edit: Export/Import: Wrap Cell Content: |            |           |          |              |          |            |
|                                         | EmployeeID | FirstName | LastName | DepartmentID | Salary   | JoinDate   |
| ▶                                       | 1          | Ravi      | Kumar    | 101          | 50000.00 | 2023-01-15 |
|                                         | 2          | Priya     | Sharma   | 102          | 60000.00 | 2022-05-10 |
|                                         | 3          | Rahul     | Reddy    | 101          | 55000.00 | 2024-02-20 |
|                                         | 4          | Anitha    | Rao      | 103          | 65000.00 | 2025-06-20 |
|                                         | 5          | Anitha    | Rao      | 103          | 65000.00 | 2025-06-20 |
| *                                       | NULL       | NULL      | NULL     | 103          | NULL     | NULL       |